

Facial Expression to Emoji

Leveraging ML/DL for expressive emoji conversion.

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Facial Expression to Emoji Conversion using ML/DL

Mapping Emotions to Emojis with Artificial Intelligence



Facial Recognition & Landmark Detection

The ML model first identifies the face in the input image. It then pinpoints key facial landmarks (eyes, nose, mouth).



Emotion Classification

A Deep Learning network analyzes the landmark geometry to classify the primary emotion (e.g., Happy, Sad, Angry).



Emoji Mapping & Output

The classified emotion is mapped to a corresponding emoji. Happy → 😊, Sad → 😞, Angry → 😡, etc.

Result: 😊

Introduction and Motivation

Bridging Human Expression and Digital Communication

● Novel Communication

Converting facial expressions to emojis provides a new way to interpret and communicate human emotions digitally.

● Real-World Applications

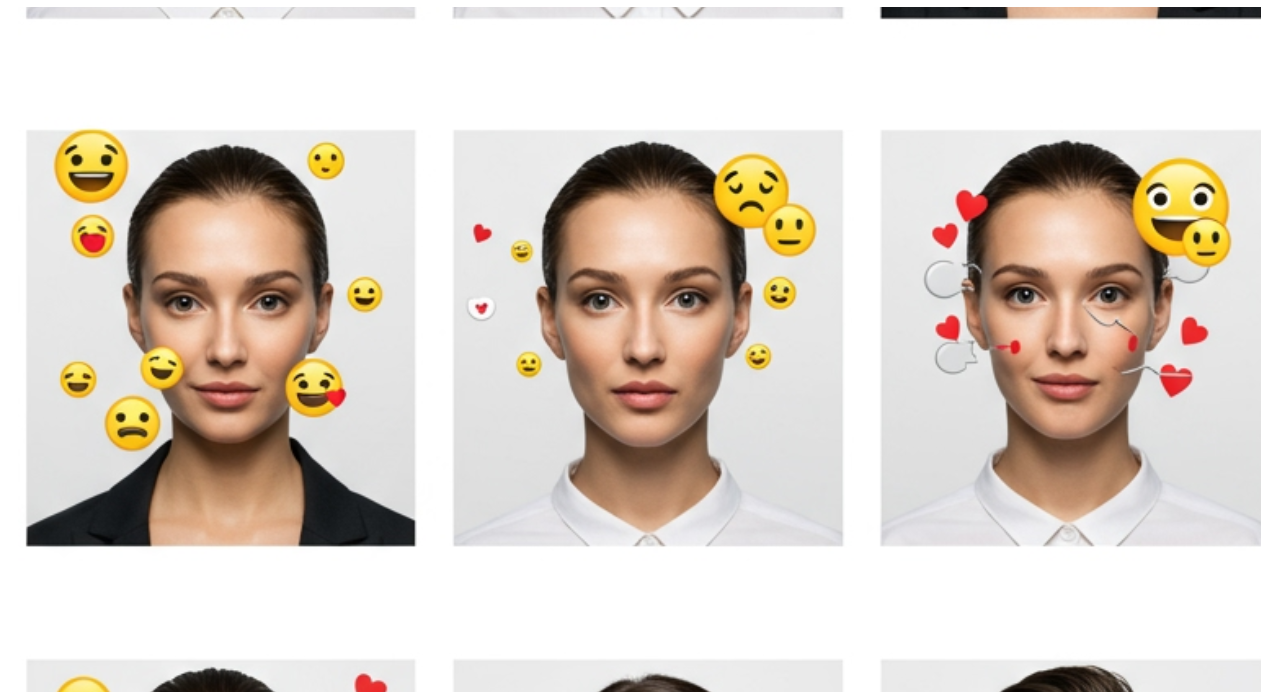
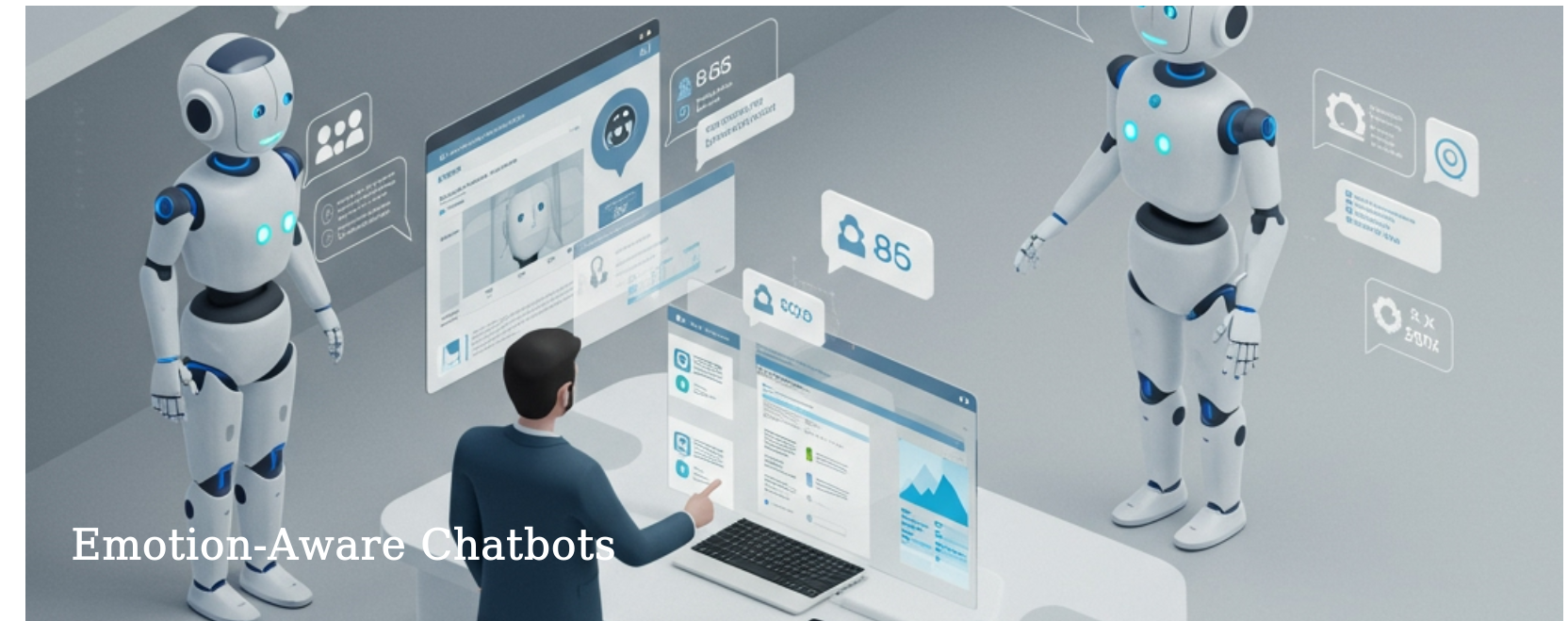
Enhancing emotion-aware chatbots for more natural interactions and creating dynamic, expression-driven social media filters.

● Mental Health & Well-being

Potential for tracking and monitoring through non-invasive emotional analysis, offering new tools for well-being applications.

● Bridging the Gap

This technology closes the divide between nuanced human expressions and their concise digital representations.



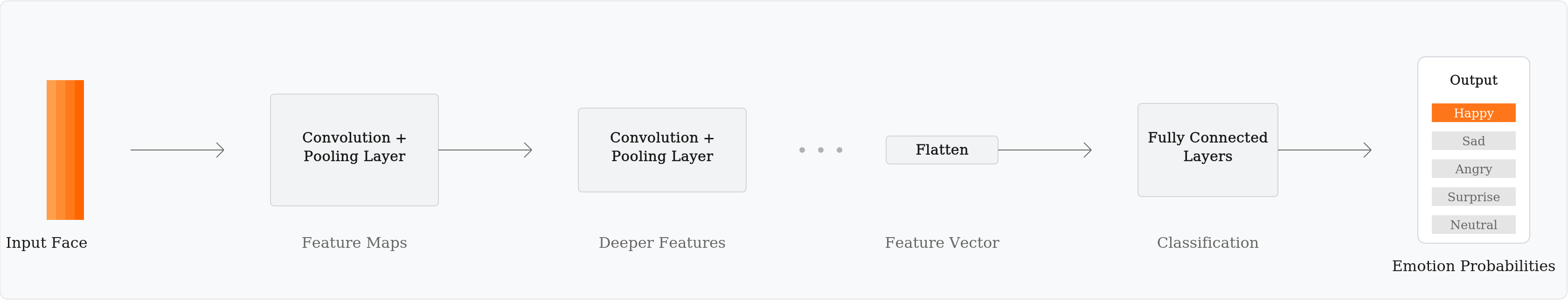
Methodology and Workflow

A step-by-step overview of the emotion detection and mapping process.

Emotion Recognition Flow



Conceptual CNN Architecture



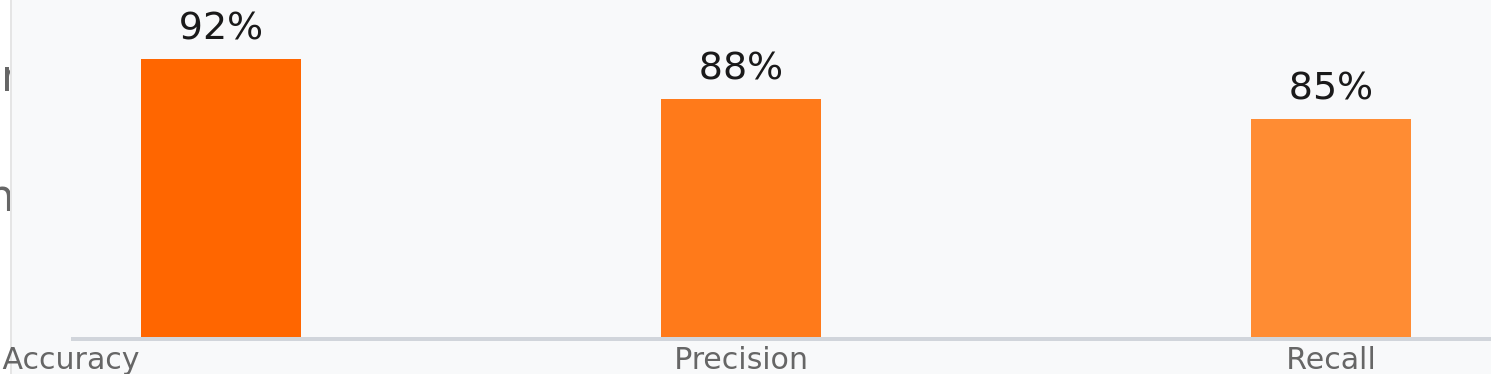
Dataset, Results, and Evaluation

Performance metrics and sample outputs for facial expression recognition.

Datasets & Frameworks

- Commonly used datasets include FER2013 and CK+ for diverse expression
- Models developed using prominent frameworks like TensorFlow or PyTorch
- Performance is assessed using Accuracy, Precision, and Recall metrics.

Model Performance Metrics



Sample Output: Expression to Emoji Mapping

The system accurately maps detected facial expressions to corresponding emojis.

Input Expression:

Sad Face (😞)

Mapped Emoji:

Crying (😭)

Input Expression:

Happy Face (😄)

Mapped Emoji:

Joy (😂)

